SR-MPLS Training Lab

M1 IS-IS Underlay

Estimated time: 60 min

Objective

Build the IS-IS Level-2 underlay that every SR-MPLS feature rides on top of. You will enable IS-IS on all transit links, confirm every adjacency reaches Up, read the link-state database, and prove loopback-to-loopback reachability. Without a correct underlay, nothing above it works.

Before you start: M0 complete: stage0-base committed on every node, so each transit interface already carries family inet, family iso, and family mpls, and each loopback has its ISO NET address.

Concept primer

This lab runs a single-level IS-IS domain, Level 2 only. Level 1 is disabled and 'wide-metrics-only' is set so the router advertises the modern wide (32-bit) metric TLVs that Segment Routing and Traffic Engineering require. All of this ships in the ISIS-PARAMETERS group, applied with apply-groups, which also marks every ge-interface point-to-point.

Each router's identity in IS-IS is its NET address, formatted 49.0001.<loopback-in-hex>.00. PE1's loopback 10.0.0.1 becomes 49.0001.0100.0000.0001.00. The loopback interface is placed in IS-IS as 'passive' so its /32 is advertised as a reachable prefix but no adjacency is ever attempted over it; that /32 is what SR node-SIDs and BGP next-hops resolve to.

'traffic-engineering l3-unicast-topology' and 'advertisement always' populate the TE database from the regular unicast topology and keep it advertised even without an RSVP session. Later modules (Flex-Algo, TI-LFA) read that TE database. IS-IS chooses paths by lowest cumulative metric, not hop count, so the metrics you set here decide which plane traffic takes. Every core link is metric 10; the P1-P2 and P3-P4 cross-links are metric 30 so they are used only as backups.

Build steps

1. **Apply the ISIS-PARAMETERS group and enable the TE topology.**

```
set groups ISIS-PARAMETERS protocols isis interface <ge-*> point-to-point
set groups ISIS-PARAMETERS protocols isis level 1 disable
set groups ISIS-PARAMETERS protocols isis level 2 wide-metrics-only
set apply-groups ISIS-PARAMETERS
set protocols isis traffic-engineering l3-unicast-topology
set protocols isis traffic-engineering advertisement always
```

Note: These six lines are immutable mandates. They must appear on every PE, core, and transit node.

2. **Make the loopback passive so its /32 is advertised, not adjacent.**

```
set protocols isis interface lo0.0 passive
```

3. **Enable IS-IS on each transit interface with its metric.**

```
set protocols isis interface ge-0/0/1.0
set protocols isis interface ge-0/0/1.0 level 2 metric 10
set protocols isis interface ge-0/0/2.0
set protocols isis interface ge-0/0/2.0 level 2 metric 10
commit and-quit
```

Note: PE1's two transit links are ge-0/0/1 (toward P1, plane A) and ge-0/0/2 (toward P2, plane B). Each node's interfaces and metrics are in its own configs/stage1-isis/<node>.set. Repeat per node, or push the whole stage with 'python push_lab_configs.py stage1-isis'.

Verification

show isis adjacency

```
Interface             System   L State   Hold (secs)
ge-0/0/1.0            P1       2  Up          23
ge-0/0/2.0            P2       2  Up          19
```

What to look for: Every expected neighbor is Up. PE1 and PE2 have two neighbors each; P and C routers have two or three. A Down or Init state points at a physical link, MTU, or family-iso problem, not a metric problem.

show isis interface

```
Interface      L  CirID Level 1 DR   Level 2 DR   L1/L2 Metric
lo0.0          0  0x1   Passive      Passive       0/0
ge-0/0/1.0     2  0x1   Disabled     Point to Point  10/10
```

What to look for: lo0.0 is Passive; transit links are Point to Point at metric 10. A wrong metric here is exactly the class of fault in this module's break/fix.

show isis database

```
IS-IS level 2 link-state database:
LSP ID        Sequence  Checksum  Lifetime
PE1.00-00     0x12      0x8a3c    1094
PE2.00-00     0x0f      0x77b1    1101
P1.00-00 ... C1.00-00 ... (all 8 SR nodes present)
```

What to look for: All eight SR nodes (PE1, PE2, C1, C2, P1-P4) appear in the database. A missing LSP means that node has no adjacency into the domain.

ping 10.0.0.2 source 10.0.0.1 rapid

```
PING 10.0.0.2 (10.0.0.2): 56 data bytes
!!!!!
round-trip min/avg/max = 1.2/2.0/3.1 ms
```

What to look for: PE1's loopback reaches PE2's loopback through the IS-IS underlay. This is the plain-IP reachability that M2 will move onto MPLS label-switched paths.

[screenshot 002: M1 verification: IS-IS adjacencies Up and full database]

Break / Fix exercise

Scenario: IS-IS is up and PE1 can reach PE2's loopback, but a traceroute shows traffic taking the plane-B path (PE1-P2-C2-P4-PE2) when the design says plane A should be equal-cost. Every adjacency reads Up, so nothing looks wrong at first glance.

Your task: Find why SPF prefers the wrong plane and restore equal-cost balance across plane A and plane B, using only IS-IS metric configuration.

Hint: Adjacency Up tells you nothing about metrics. Compare the advertised metric on each plane-A link in the database against the design value of 10, and check the total cost of 'show route 10.0.0.2' against the expected 40.

The instructor holds the fault-injection and fix commands in the answer key. Diagnose it yourself first.

Brief it back

Explain each of these aloud before moving on. If you cannot, re-read the primer and re-run the verification commands.

- Explain how IS-IS builds its SPF tree by cumulative metric rather than hop count, and how that decides plane A versus plane B in this topology.

- Why is the loopback configured as passive instead of as a normal IS-IS interface?
- What does 'traffic-engineering l3-unicast-topology' add that plain IS-IS routing does not, and which later modules depend on it?